Abhishek Sudarshan Pathania

System Development Engineer 2 @ Amazon

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SUMMARY

System Development Engineer 2 at Amazon specializing in distributed event-driven systems on AWS. Deep expertise in high-concurrency Kotlin/Java architectures, Step Functions orchestration, bounded concurrency, single-retry ownership per failure class, zero-trust security migrations, and automated safe continuous delivery.

EXPERIENCE

System Development Engineer 2 — Amazon

2021 – Present | India

- **Zero-Trust Identity & Storage Migration:** Designed and executed phased migration to identity-bound OIDC authenticated access across ingestion fleets with server-side tenant isolation, safe retries, and bounded memory caps; achieved zero pre-production errors and eliminated silent data corruption.
- **Serverless Fan-Out & Cold Start Optimization:** Profiled and tuned JVM Lambda cold starts from 8–11s to ~1.5s (P99 latency −70%); implemented claim-check pattern on S3/SQS for large payloads and redesigned retry budgets to absorb dependency throttling without quota increases.
- **Self-Healing Workflows & Idempotent Recovery:** Instituted single-retry ownership per failure class with stall alarms for multi-day Step Functions workflows; built idempotent bulk-recovery CLI (redrive, step restart, dry-run) slashing recovery from 6–8 hours to 10–15 minutes (MTTR −70%, ~15% compute saved on EC2 Spot).
- **Scalable Data Access & Cursor Pagination:** Designed backward-compatible `search_after` cursor pagination across OpenSearch indexes with sort tie-breakers; enforced TOCTOU-safe conditional writes in DynamoDB (~65% faster lookups, zero integrity failures).
- **Progressive Delivery & Production Safety:** Established team-wide deployment standards (bake times, composite alarms, automated rollback, CodeDeploy gradual shifting, cross-account synthetic canaries); raised continuous deployment from 48% to 84% and cut production lead time from 53 to 16 days.
- **AI & LLM Engineering:** Architected engineering RAG assistant on Amazon Bedrock Knowledge Bases with intent-based model routing; elevated question resolution accuracy from ~60% to ~95% across 25+ adopted teams.

Software Engineer (TCS Digital) — Tata Consultancy Services

June 2019 – 2021 | India

- **Enterprise Banking Modernization:** Delivered Proof-of-Concepts (POCs) for tier-1 European banking client modernizing legacy architectures to scalable Spring Boot microservices and Liferay DXP.
- **Data Migration Engine:** Built high-concurrency batch data migration modules leveraging client REST APIs to orchestrate cross-branch user account transfers with schema reconciliation and zero data loss.

SELECTED SYSTEMS & ARCHITECTURAL CASE STUDIES

Zero-Trust Access to Stored Assets

`Zero Pre-Prod Errors • Tenant Isolation`

Replaced replayable pre-signed URLs with identity-bound OIDC auth, KMS encryption contexts, safe retries, and bounded memory caps on untrusted inputs.

Serverless Event-Driven Fan-Out at Scale

`Cold Start 8–11s → ~1.5s • P99 −70%`

Queue fan-out with claim-check pattern for large messages, provisioned concurrency tuning, and CodeDeploy gradual traffic shifting with composite alarm rollback.

Self-Healing Multi-Day Workflows & Bulk Recovery CLI

`MTTR −70% • Recovery 6–8h → 10–15m`

Orchestrated Step Functions with single retry owners, stall detection alarms, and an idempotent CLI for dry-run validation and state-aware bulk redrive.

High-Throughput Cursor-Paginated History API

`~65% Faster Lookups • TOCTOU-Safe`

Backward-compatible cursor pagination contract with sort tie-breakers across multiple search indexes; conditional writes in DynamoDB for race safety.

SKILLS & COMPETENCIES

Distributed Systems:	Event-driven fan-out, claim-check, jittered exponential backoff, Step Functions, token bucket rate limiters, multi-tenant OIDC isolation
AWS Cloud Platform:	Lambda, ECS Fargate, EC2 Spot, SQS, SNS, Step Functions, S3, API Gateway, DynamoDB, OpenSearch, KMS, IAM, VPC PrivateLink

Languages & Core:	Kotlin (primary), Java, TypeScript, Python, Smithy API models, Concurrency Control, Memory Caps, Coroutines, MockK, JUnit 5
AI & LLM Engineering:	Amazon Bedrock Knowledge Bases (RAG), Converse API, Intent Classification, Structured Outputs, Strands Agents, MCP Servers
DevOps & Delivery:	AWS CDK (TypeScript), CloudFormation, CodeDeploy Gradual Shifting, Composite Alarms, Synthetic Canaries, Docker, Primary On-Call

EDUCATION

Bachelor of Science in Computer Science & Engineering — Gujarat Technological University 2015 – 2019 | India